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1 Direct Limits

Suppose $M_i = N_i \oplus P_i$ and $\phi_{i',i} = \psi_{i',i} + \rho_{i',i}$. Let N, P be the direct limits of $N = \lim_{\rightarrow} N_i$ and $P = \lim_{\rightarrow} P_i$, so we have induced $\psi : N \rightarrow M$ and $\rho : P \rightarrow M$, so $\psi\psi_i = \phi_i|M_i, \rho\rho_i = \phi_i|P_i$. Then $\psi \oplus \rho : N \oplus P \rightarrow M$ is an isomorphism.

First, I claim $\psi_{i'',i'}\psi_{i',i} = \psi_{i'',i}$ and similarly for ρ . Write $\phi_{i'',i'}\phi_{i',i} = \phi_{i'',i}$, so $(\psi_{i'',i'} + \rho_{i'',i'})(\psi_{i',i} + \rho_{i',i}) = \psi_{i'',i} + \rho_{i'',i}$. Expanding and matching gives $\psi_{i'',i'}\psi_{i',i} = \psi_{i'',i}$. I claim that ψ, ρ can be constructed. We check $\psi_{i'}\psi_{i',i} = \phi_{i'}\phi_{i',i} = \phi_i = \psi_i$, so ψ is defined.

Define $\theta : M \rightarrow N \oplus P$ by $\theta(x) = (\psi_i y_i, \rho_i z_i)$ if $x = \phi_i x_i$ and $x_i = y_i + z_i$. I claim that θx is independent of i . Suppose $x = \phi_i(x_i) = \phi_{i'}(x_{i'})$. Pick $i'' \geq i, i'$. Then $x = \phi_i(x_i) = \phi_{i''}\phi_{i'',i}(x_i) = \phi_{i''}(\psi_{i'',i}y_i + \rho_{i'',i}z_i) = \psi_i y_i + \rho_i z_i$. We have a similar equation for i' , so matching gives $(\psi_i y_i, \rho_i z_i) = (\psi_{i'} y_{i'}, \rho_{i'} z_{i'})$. We can check that θ is a homomorphism.

Suppose $A'_i \xrightarrow{\mu_i} A_i \xrightarrow{\varepsilon_i} A''_i$ is exact for all i and we have the following (c) diagram:

$$\begin{array}{ccccc} A'_i & \xrightarrow{\mu_i} & A_i & \xrightarrow{\varepsilon_i} & A''_i \\ \downarrow \phi'_{i',i} & & \downarrow \phi_{i',i} & & \downarrow \phi''_{i',i} \\ A'_{i'} & \xrightarrow{\mu_{i'}} & A_{i'} & \xrightarrow{\varepsilon_{i'}} & A''_{i'} \end{array}$$

Then $A' \xrightarrow{\mu} A \xrightarrow{\varepsilon} A''$ is exact.

Proof: Suppose $\varepsilon a = 0$ for some $a \in A$. Let $a = \phi_i a_i$, so $\varepsilon a = \varepsilon \phi_i a_i = \varepsilon_i a_i = 0$, so $a_i = \mu_i a'_i$. Then $a = \phi_i \mu_i a'_i = \mu \phi'_i a'_i$. I claim that the direct limit is “functorial.” Suppose $A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i$ with commuting and $A \xrightarrow{f} B \xrightarrow{g} C$. Then $gf = \lim_{\rightarrow} (g_i f_i)$. Indeed, pick $a = \phi_i a_i$. Then $gfa = g(\phi_i f_i a_i) = \rho_i g_i f_i a_i = \lim_{\rightarrow} (g_i f_i)(a)$. It follows that $\varepsilon \mu = 0$.

As a corollary, if $f_i : A_i \rightarrow B_i$ are injective (surjective), then so is $f : A \rightarrow B$.

Let $J \leq I$ be final. Define $\lambda : \lim_{\rightarrow} M_j \rightarrow \lim_{\rightarrow} M_i$. I claim that λ is an isomorphism.

Proof: Let $\phi_j : M_j \rightarrow \lim_{\rightarrow} M_j$ and $\varphi_i : M_i \rightarrow \lim_{\rightarrow} M_i$. Suppose $y \in \lim_{\rightarrow} M_i$, so $y = \varphi_i y_i$. Pick $j \geq i$, so $y = \varphi_j \varphi_{j,i} y_i = \varphi_j y_j$ for $y_j = \varphi_{j,i} y_i$. Then $y = \lambda(\phi_j y_j)$, so λ is surjective. Suppose $\lambda x = 0$, so $x = \phi_j x_j$, so $\lambda x = \varphi_j x_j$. If $\varphi_j x_j = 0$, show $\phi_j x_j = 0$. Write N_J to be the submodule of $\bigoplus_{j \in J} M_j$ generated by $\iota_{j'} \phi_{j',j} x_j - \iota_j x_j$ for all $j \leq j'$ and $x_j \in M_j$. Denote by N_I similarly.

We have $\iota_j x_j \in N_J$. Write $\iota_j x_j = \sum_{k=1}^n (\iota_{j'_k} \phi_{j'_k,j_k} x_{j_k} - \iota_{j_k} x_{j_k})$. There exists a homomorphism $\bigoplus_{j \in J} M_j \rightarrow \bigoplus_{i \in I} M_i$ by sending elements of index $j \in J$ to $j \in I$. Applying the homomorphism, we have $\kappa_j x_j = \sum_{k=1}^n (\kappa_{j'_k} \phi_{j'_k,j_k} x_{j_k} - \kappa_{j_k} x_{j_k})$, so $\phi_j x_j = 0$.

Suppose $\phi_{i',i} : M_i \rightarrow M_{i'}$ is an isom for all $i \leq i'$. Then $\phi_i : M_i \rightarrow M$ is an isom for all i .

Proof: I claim that ϕ_i is surjective. Let $x \in M$, so $x = \phi_{i_1}(x_{i_1})$ for some i_1 . Pick $i_2 \geq i, i_1$, so $x = \phi_{i_2} \phi_{i_2,i_1}(x_{i_1}) = \phi_{i_2} \phi_{i_2,i} \phi_{i_2,i_1}^{-1}(x_{i_1}) = \phi_i \phi_{i_2,i}^{-1} \phi_{i_2,i_1}(x_{i_1})$. I claim that ϕ_i is injective. Suppose $\phi_i x_i = 0$, so $\iota_i x_i = \sum_{(j',j)} \iota_{j'} \phi_{j',j}(x_{j',j}) - \iota_j(x_{j',j})$, so $x_i = \sum_{(i,j)} \phi_{i,j} x_{i,j} - \sum_{(j',i)} x_{j',i}$ and $0 = \sum_{(i',j)} \phi_{i',j}(x_{i',j}) - \sum_{(j',i')} x_{j',i'}$. Pick $k \geq j', j$ larger than all of the j', j . Then

$$\begin{aligned} \phi_{k,i} x_i &= \sum_{(i,j)} \phi_{k,j} x_{i,j} - \sum_{(j',i)} \phi_{k,i} x_{j',i} + \sum_{(i',j)} \phi_{k,j} x_{i',j} - \sum_{(j',i')} \phi_{k,j'} x_{j',i'} \\ &= \sum_{(j',j)} \phi_{k,j} x_{i,j} - \sum_{(j',j)} \phi_{k,i} x_{j',j} = 0, \end{aligned}$$

so $x_i = 0$.

Claim: Let $I', I'' \subseteq I$ and $I' \leq I''$. Let M', M'' denote the limits over I', I'' . Pick $i' \in I'$ and $i'' \in I''$ greater than i' . Let $\phi'_{i'} : M_{i'} \rightarrow M'$ and $\phi''_{i''} : M_{i''} \rightarrow M''$. If $\phi'_{i'} x_{i'} = 0$, then $\phi''_{i''} \phi_{i'',i'} x_{i'} = 0$.

Proof: There exists $j_1 \geq i'$ in I' such that $\phi_{j_1,i'} x_{i'} = 0$. Pick $j_2 \geq j_1$ in I'' . Pick $j_3 \geq j_2, i''$ in I'' . Then

$$\begin{aligned} \phi''_{i''} \phi_{i'',i'} x_{i'} &= \phi''_{j_3} \phi_{j_3,i''} \phi_{i'',i'} x_{i'} = \phi''_{j_3} \phi_{j_3,i'} x_{i'} \\ &= \phi''_{j_3} \phi_{j_3,j_1} \phi_{j_1,i'} x_{i'} = \phi''_{j_3} \phi_{j_3,j_1} 0 = 0 \end{aligned}$$

Claim: Let J be another directed set and for each $j \in J$, associate $I_j \leq I$ such that if $j \leq j'$, then $I_j \leq I_{j'}$. Further, assume $I = \bigcup_{j \in J} I_j$. Define $M_j^* = \lim_{i \in I_j} M_i$. Then $\lim_{j \in J} M_j^* = M^*$.

Proof: First, we show if $j \leq j'$, we have the associated $\psi_{j',j} : M_j^* \rightarrow M_{j'}^*$. Suppose $x \in M_j^*$, so $x = \phi_{i_1}^*(x_{i_1})$. Pick $i_1 \leq i_2$ for $i_2 \in I_{j'}$. Define $\psi_{j',j}x = \phi_{i_2}^*\phi_{i_2,i_1}(x_{i_1})$. Pick i_3, i_4 similarly. Suppose $\phi_{i_1}^*(x_{i_1}) = \phi_{i_3}^*(x_{i_3})$. Pick $i_5 \geq i_1, i_3$ in I_j . Then $\phi_{i_5}^*\phi_{i_5,i_1}x_{i_1} = \phi_{i_5}^*\phi_{i_5,i_3}x_{i_3}$, so $\phi_{i_5}^*(\phi_{i_5,i_1}x_{i_1} - \phi_{i_5,i_3}x_{i_3}) = 0$. Pick $i_6 \geq i_2, i_4, i_5$ in $I_{j'}$. From the previous result, we have $\phi_{i_6}^*\phi_{i_6,i_5}(\phi_{i_5,i_1}x_{i_1} - \phi_{i_5,i_3}x_{i_3}) = \phi_{i_6}^*\phi_{i_6,i_1}x_{i_1} - \phi_{i_6}^*\phi_{i_6,i_3}x_{i_3} = \phi_{i_6}^*\phi_{i_6,i_2}\phi_{i_2,i_1}x_{i_1} - \phi_{i_6}^*\phi_{i_6,i_4}\phi_{i_4,i_3}x_{i_3} = \phi_{i_2}^*\phi_{i_2,i_1}x_{i_1} - \phi_{i_4}^*\phi_{i_4,i_3}x_{i_3} = 0$. Thus, $\psi_{j',j}$ is well-defined.

Let $M' = \lim_{j \in J} M_j^*$. Define $\varphi_i : M_i \rightarrow M'$ by $\varphi_i x_i = \psi_j \phi_i^* x_i$ for some j such that $i \in I_j$. I claim that $\varphi_i x_i$ is independent of j .

Claim: Let $\{M_j, \phi_{j',j}\}$ be a directed system and $\{M, \phi_j\}$ a module with homs $\phi_j : M_j \rightarrow M$ such that $\phi_{j'}\phi_{j',j} = \phi_j$ for all $j \leq j'$. Suppose further $M = \bigcup_{j \in J} \text{im } \phi_j$. Then $\{M_j, \phi_j\}$ is the associated direct limit of $\{M_j, \phi_{j',j}\}$.

Proof: Let $\psi_j : M_j \rightarrow N$ such that $\psi_{j'}\phi_{j',j} = \psi_j$. We wish to show there exists a unique $\psi : M \rightarrow N$ such that $\psi\phi_j = \psi_j$. Suppose $x \in M$. Pick j_1 such that $x = \phi_{j_1}x_{j_1}$ and define $\psi x = \psi_{j_1}x_{j_1}$. I claim that ψ is well-defined. We have $\phi_{j_1}x_{j_1} = \phi_{j_2}x_{j_2}$. Pick $j_3 \geq j_1, j_2$, so $\phi_{j_3}(\phi_{j_3,j_1}x_{j_1} - \phi_{j_3,j_2}x_{j_2}) = 0$. Then $\psi_{j_3}(\phi_{j_3,j_1}x_{j_1} - \phi_{j_3,j_2}x_{j_2}) = \psi_{j_1}x_{j_1} - \psi_{j_2}x_{j_2} = 0$, so ψ is well-defined. Further, ψ is unique: suppose ψ, ψ' satisfy the given property and $x \in M$. Then $\psi x = \psi_j x_j = \psi' x$. Thus, $M = \lim_{j \in J} M_j$.

Claim: Let X be Hausdorff. Then $\lim_{U \in \mathcal{U}} H_n(X, X - U) = H_n(X, X - x)$, where $\mathcal{U} = \{U : x \in U \text{ and } U \text{ is open}\}$.

Proof: Note the partial order on $\mathcal{U}$ is given by $U \preceq V$ if $U \geq V$. The maps are given by $j_V^U : H_n(X, X - U) \rightarrow H_n(X, X - V)$ (which form a direct system) and $j_x^U : H_n(X, X - U) \rightarrow H_n(X, X - x)$. We have $j_x^V j_V^U = j_x^U$. Suppose $\bar{z} \in H_n(X, X - x)$. Let $U = X - |\partial z|$, which is open (since $|\partial z|$ is compact and X is Hausdorff), and contains x (since $|\partial z| \leq X - x$). Thus, we have the inclusion $j_x^U(\bar{z}) = \bar{z}$, where the inner bar is taken over $H_n(X, X - U)$. It follows that $H_n(X, X - x) = \bigcup_{U \in \mathcal{U}} \text{im } j_x^U$, so the limit follows from the above claim.

2 Orientation of Manifolds Revisited

Let $\mathcal{U}$ be a covering of X such that j_x^U is an isomorphism for all $x \in U$. A collection $\{(U, \alpha_U) : U \in \mathcal{U}, H_n(X, X - U) = \langle \alpha_U \rangle\}$ is said to be an R -orientation system if for all $x \in U \cap V$, that $j_x^U(\alpha_U) = j_x^V(\alpha_V) = \alpha_x$. The local orientation at x is the element α_x .

Claim: S^n is orientable.

Proof: $n = 0$ works, so assume $n > 0$. Take α to be a generator of $H_n(S^n)$. I claim that $\{(S^n, \alpha)\}$ is an R -orientation system. Compatibility is done. We check

$$\begin{aligned} H_n(S^n - x) &\rightarrow H_n(S^n) \xrightarrow{j_x^X} H_n(S^n, S^n - x) \rightarrow H_{n-1}^\#(S^n - x) \\ 0 &\rightarrow H_n(S^n) \xrightarrow{j_x^X} H_n(S^n, S^n - x) \rightarrow 0, \end{aligned}$$

so j_x^X is an isomorphism (we use $S^n - x \simeq \mathbb{R}^n \simeq 0$).

R -orientation sheaf. Let $X^0 = \{(x, \alpha_x) : \alpha_x \in H_n(X, X - x)\}$. Give X^0 the basis $\mathcal{B}$ of sets $\langle U, \alpha_U \rangle = \{(x, \alpha_x) : x \in U, \alpha_x = j_x^U(\alpha_U), j_x^U \text{ is an isomorphism for all } x \in U\}$.

Claim: Let R be an integral domain and $M \cong R$ with the isomorphism over R -mod. Suppose $M = \langle x_0 \rangle = \langle y_0 \rangle$ over R -mod. Then $y_0 = ux_0$ for some $u \in R^\times$.

Proof: There exists $u \in R$ such that $y_0 = ux_0$. Pick $v \in R$ such that $x_0 = vy_0$. Then $x_0 = vux_0$. Let $\varphi : R \rightarrow M$ be the given isom with $\varphi(1) = z_0$. Suppose $rx_0 = 0$ for some $r \in R$. Pick s such that $x_0 = sz_0$, so $rsz_0 = rs\varphi(1) = \varphi(rs) = 0$, so $rs = 0$. If $r = 0$, we're done. Else, $s = 0$, so $x_0 = 0$, absurd (we assume $R \neq 0$). Thus, $(vu - 1)x_0 = 0$, so $vu = uv = 1$ (we use R is commutative, and has identity).

Hence, we define the valuation $v : X^0 \rightarrow R/R^\times$ by $v(x, \beta) = \bar{r}$, where $\beta_x = r\alpha_x$ for some generator α_x of $H_n(X, X - x)$. I claim that if $R/R^\times$ is given the discrete top, then v is continuous.

Proof: Suppose $v(x, \beta_x) = \bar{r}$. Pick the generator α_x such that $\beta_x = r\alpha_x$. Let $U(x)$ be such that j_y^U is an isom for all $y \in U$ and α_U is the local continuation of α_x over U . I claim that $v(\langle U, r\alpha_U \rangle) = \bar{r}$. Then $j_y^U(r\alpha_U) = rj_y^U\alpha_U$. Since j_y^U is an isom, we have that $j_y^U(\alpha_U)$ is a generator. Thus, the result follows.

Claim: Let $s : X \rightarrow X^0$ be a global section and $x \in X$. Then there exists special $U(x)$ such that $s(U) = \langle U, \alpha_U \rangle$ for some $\alpha_U \in H_n(X, X - U)$.

Proof: Let $(x, s'(x)) \in \langle V, \alpha_V \rangle$ for special $V(x)$. Since s is continuous, pick $W(x)$ such that $s(W) \leq \langle V, \alpha_V \rangle$, so $W \leq V$ and $j_y^V(\alpha_V) = s'(y)$ for all $y \in W$. Pick special $U(x) \leq V \cap W$ and let $\alpha_U = (j_x^U)^{-1}(s'(x))$. Then for all $y \in U$,

$$\begin{aligned} j_y^U(\alpha_U) &= j_y^U(j_x^U)^{-1}(s'(x)) = j_y^U(j_x^U)^{-1}j_x^V(\alpha_V) \\ &= j_y^U j_U^V(\alpha_V) = j_y^V(\alpha_V) = s'(y), \end{aligned}$$

so indeed $s'(U) = \langle U, \alpha_U \rangle$.

Claim: X is R -orientable iff there exists a section $s : X \rightarrow v^{-1}(1)$.

Proof: Let $X' = \text{im } s$. Let $\{(U, \alpha_U)\}$ be an exact cover of X' (since X' is open). I claim that $\{(U, \alpha_U)\}$ is an R -orientation system of X . Indeed, the U 's cover X and if $(U, \alpha_U), (V, \alpha_V)$ are in the system and $x \in U \cap V$, then

$$j_x^U(\alpha_U) = s'(x) = j_x^V(\alpha_V).$$

The elements α_U are generators since $v(x, \alpha_U) = \bar{1}$.

An orientation along A is a section $s : A \rightarrow v^{-1}(1)$. Claim: X is R -orientable along A iff there exists a homeomorphism $\Phi : p^{-1}A \rightarrow A \times R$ such that the following diagram commutes:

$$\begin{array}{ccc} p^{-1}A & \xrightarrow{\Phi} & A \times R \\ & \searrow & \swarrow \\ & A & \end{array}$$

where R is given the discrete topology.

When can we extend a section $s : A \rightarrow X^0$ to an open set containing A ? Let $U \leq X$ be special and pick $\alpha_U \in H_n(X, X - U)$ and define $s : U \rightarrow X^0$ by $s(x) = j_x^U(\alpha_U)$. Then s is continuous.

Proof: Suppose $s(y) = j_y^U(\alpha_U) \in \langle V, \alpha_V \rangle$ for some special V . Pick $W(x) \leq U \cap V$ special. Then for $y \in W$,

$$\begin{aligned} s(y) &= j_y^U(\alpha_U) = j_y^W j_W^U (j_x^W j_W^U)^{-1} j_x^V \alpha_V \\ &= j_y^W j_W^U (j_W^U)^{-1} (j_x^W)^{-1} j_x^W j_W^V (\alpha_V) \\ &= j_y^V(\alpha_V), \end{aligned}$$

so $s(W) \leq \langle V, \alpha_V \rangle$.

Prove: Connect sum is well-defined (gives manifold).

Check 25.17?

If M is finitely generated over pid R , then $\text{Ext}_R(M, R) = T(M)$.

Define $d : H^q(A) \rightarrow H^{q+1}(X, A)$ as follows. Suppose $\bar{c} \in H^q(A)$ and $\bar{z} \in S_{q+1}(X, A)$. Let $d(\bar{c})(\bar{z}) = [\partial z, c]$, where $\bar{c}$ is a cocycle and $\bar{z}$ is a relative (X, A) cochain. First, we show that d is well-defined. Suppose $c_2 - c_1 = \delta c'$ and $z_2 - z_1 = z'$ for some cochain c' and A -chain z' . Then

$$\begin{aligned} [\partial(z_1 + z'), c_1 + \delta c'] &= [\partial z_1 + \partial z', c_1 + \delta c'] = [\partial z_1 + \partial z', c_1] + [\partial z_1 + \partial z', \delta c'] \\ &= [\partial z_1, c_1] + [\partial z', c_1] = [\partial z_1, c_1] + [z', \delta c_1] = [\partial z_1, c_1], \end{aligned}$$

so d is indeed well-defined. Now, we show that indeed $d(\bar{c})$ is a cocycle. Let $\bar{z} \in S_{q+2}(X, A)$. Then

$$[\bar{z}, \delta d(\bar{c})] = [\partial^2 z, c] = 0,$$

so $d(\bar{c}) \in Z^{q+1}(X, A)$.